INSTRUMENTATION & ELECTRICAL











NETA

TEL: 01642 616936

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TAILORED COURSES ALSO AVAILABLE

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Introduction

NETA is recognised as one of the UK's leading providers of vocational skills training and is an award winning company involved in the training and re-skilling of both young people and adults.

Training methods are competency based and all NETA's instructional staff have come from industry with considerable personal experience within their field of expertise.

The philosophy of the company, which is central to all its services, is that individuals should receive high quality training in an environment, which simulates as far as possible, the real working conditions found in industry. The integration of working rigs and simulated site activity into the learning process maximises the effective use of resources and creates a realistic learning experience for individuals.

A major strength has been the ability of the company to develop learning programmes, which meet specific client requirements rather than limiting choice to standard course activity.

In order to implement this, NETA has, over the years, developed a number of well-equipped training centres in the Tees Valley in Northern England. In addition, many courses are now delivered at clients' own premises either in the UK or abroad, and the majority of NETA's programmes are customised, to reflect specific requirements, offering a flexibility that guarantees clients the best training delivered in the most efficient and cost effective way.

This booklet details the range of standard Instrument and Electrical courses currently on offer, however NETA can tailor training to clients exact requirements. More information is available on request.

5 Days Basic Electrical Skills for Personnel of other Disciplines

Course Reference: CP - 8049

Aim

The aim of this course is to give basic Electrical Skills to Personnel of other disciplines

Pre-Requisite

Delegates will ideally be from a craft engineering discipline

Course Objectives

On completion of this course delegates will be able to:

- Understand basic electrical theory and terminology
- Understand the dangers of electricity (shock, burns etc)
- Understand their responsibilities under the Electricity at Work Regulations 1989
- Understand the principles of earthing and protection systems
- Identify the safety requirements for safe isolation and de-isolation
- Understand and identify typical electrical components found within a motor control panel
- Understand the general principles and construction of AC motors, and identify three-phase and single phase machines.
- Use a Continuity and Insulation Resistance tester to test motor windings.
- Use Voltage Indicators and Proving Unit to test and prove for dead.
- Make off cable terminations correctly using SWA glands.
- Understand the principles of a direct–on–line motor control.
- Identify the safety requirements for the isolation and de-isolation of an A.C. induction motor

Course Duration

The duration of this course is five days.

Certification

Delegates will receive a NETA certificate on successful completion of the course

10 Days Basic Electrical Skills for Personnel of Other Disciplines

Course Reference: CP - 8049a

Aim

The aim of this course is to give Basic Electrical Skills to Personnel of other disciplines.

Pre-Requisite

Delegates will ideally be from a craft engineering discipline

Course Objectives

On completion of Part A the delegates will be able to:

- Work safely on electrical equipment within the scope of the Electricity at Work Regulations 1989.
- Understand and interpret Ohm's Law and solve problems relating to the Law.
- Understand and identify rewireable fuses, RCD's and circuit breakers, and be able to select the correct device to satisfy the IEE Regulations.
- Make up a range of plugs, leads and sockets to correct Industrial/Domestic standards.
- Use electrical test equipment safely and correctly, including multimeters and insulation testers - P.A.T's etc.
- Understand the importance of The IEE Regulations, cable sizing and current rating.
- Make off cable terminations correctly for industrial and domestic electrical applications, including S.W.A. and M.I.C.C. glands.

On completion of Part B the delegates will be able to:-

- Work safely on electrical equipment within the scope of the Electricity at Work Regulations 1989.
- Understand electro-magnetic induction and principles relating to electrical machines. The simple alternator.
- Understand the general construction of AC motors, and test windings and insulation of 3-phase and single-phase machines.
- Construct and understand control circuits including Softstart D.O.L. and Star/Delta starting and sequential motor control.
- Carry out a safe isolation of electrical machinery.

Course Duration

The course is in two parts. Part A - 5 days Part B - 5 days

Certification

Delegates will receive a NETA certificate on successful completion of the course

Electricity at Work Regulations 1989

Course Reference: CP - 8087

Aim

The aim of this course is to provide Electricians/Supervisors and other interested parties with the required knowledge to assume their duties and responsibilities under the Electricity at Work Regulations 1989.

Pre- Requisite

No previous knowledge of the Electricity at Work Regulations is required.

Course Objectives

Upon completion of this training course, delegates will be able to:

- Understand the requirements of the Electricity at Work Regulations
- Apply the principles of the of the Electricity at Work Regulations in a working environment

Course Duration

The course will be of one day duration.

Certification

Delegates will receive a NETA certificate on successful completion of the course

Electrical Legislation and Safety for Domestic Dwellings

Course Reference: CP - 8207

Aim

The aim of this course is to provide an awareness of a Landlord's responsibilities in respect of current legislation with reference to electrical safety in domestic dwellings.

Pre- Requisite

Candidate must either be in general management or in a supervisor role of engineering contractors.

Course Objectives

On completion of the course the delegate will have an awareness of:-

- Accidents involving Electricity
- Legislation Overview
- Electricity at Work Regulations 1989
- Inspection and Testing
- Landlord's Responsibility for Electrical Safety
- Electrical Safety in the House
- Advice to Tenants
- Home Safety Advice

Course Duration

The course will be of 1 day duration.

Certification

On successful completion of this course delegates will receive a NETA certificate.

Domestic Installer Qualification
Part P

Course Reference CP - 8211

Aim

To provide training for persons carrying out electrical work in dwellings, to meet the requirements of Part P of the Building Regulations. This is an approved NICEIC (National Inspection Council for Electrical Installation Contractors) Domestic Installer Course

Pre-Requisite

Previous electrical knowledge / experience would be an advantage, but not essential

Course Objectives

On satisfactory completion of the course the delegate will be able to:-

- Understand the requirements for Part P and other relevant legislation
- Identify the basic needs for earthing, bonding and other installation considerations
- Carry out Safe Isolation procedure and proving dead
- Carry out a Pre-Work Survey on an existing installation for alterations or adding additional circuits to the installation
- Design a circuit to add to an existing installation including protection device, cable sizing, earthing and bonding
- Identify the requirements for special locations e.g. Bathroom
- Inspect an installation and identify any defects or non-compliances
- Test circuit(s) in accordance with the requirements of BS7671
- Complete certification documents for alterations and additions to an installation

To obtain registration as a domestic installer with the NICEIC, a further on-site assessment will be carried out by NICEIC

Course Duration

The duration of this course is 4 days + 1 day assessment

Certification

Delegate will receive a NICEIC certificate on completion of the course and assessment

17th Edition Wiring Regulations (BS7671) City & Guilds 2382-10

Course Reference: CP - 8208

Aim

To provide a working knowledge of the 17th Edition IEE Wiring Regulations

Pre- Requisite

Personnel with Craft, Technician or Engineer Level experience in an Electrical discipline. City & Guilds also require candidates to produce Photographic ID (passport, driving license, etc) for the examination.

Course Objectives

On completion of this training course each delegate will be capable of:-

- Understanding the requirements of the Regulations
- Checking requirements for a Standard Installation
- Understanding the general characteristics of an installation
- Complete a 2 Hour 60 Question GOLA online examination

Course Duration

The course duration will be 3 days.

Certification

Each delegate will receive a Level 3 Certificate in the 'Requirements for Electrical Installation (BS7671 2008)' issued by City & Guilds upon successful completion of GOLA on-line examination.

17th Edition Wiring Regulations (BS 7671)
City & Guilds 2382-20 UPDATE

Course Reference: CP - 8208A

Aim

To provide a working knowledge of the changes from 16th edition IEE regulations to the 17th edition IEE Regulations

Pre Requisites

Personnel Must have completed either:

- City & Guilds Level 3 Certificate in the Requirements for Electrical Installations (BS 7671: June 2001) (2381), or
- City & Guilds Level 3 Certificate in Electrotechnical Technology Installation (Building and Structures) (2330-07), or
- City & Guilds Level 3 Certificate in Electrotechnical Technology Electrical Maintenance (2330-08), or
- City & Guilds Level 3 Certificate in Electrical Installations Part Two (2360) (from 1 January 2004 onwards only).

City & Guilds require us to take copies of the above certificate(s) and confirm the identity of the candidate via Photographic ID (passport, driving license, etc) for both the examination and against previous certification.

Course Objectives

On completion of this training course each delegate will be capable of:-

- Understanding the requirements of the Regulations
- Checking requirements for a Standard Installation
- Understanding the general characteristics of an installation
- Complete a 1 Hour 30 Question GOLA online examination

Course Duration

The course duration will be 1 days.

Certification

Each delegate will receive a Level 3 Certificate in the 'Requirements for Electrical Installation (BS7671 2008)' issued by City & Guilds upon successful completion of GOLA on-line examination.

City & Guilds 2391
Inspection Testing & Certification of Electrical Installations

Course Reference: CP - 8204

Aim

This course is designed to give experienced electricians / supervisors, who are currently involved with inspection and testing, the necessary information to meet the requirements of the City and Guilds 2391 qualification

Pre-Requisite

Personnel with craft, technician or engineer level experience in an electrical discipline. Ideally delegates should have completed 17th Edition IEE Wiring Regulations Course.

Course Objectives

On successful completion of the course delegate will: -

- Understand the inspection, testing and certification requirements to meet BS7671, 17th edition IEE Wiring Regulations.
- Successfully carry out the practical assessments and written examination as required by City and Guilds, to obtain the 2391 qualification

This is an advanced course for experienced supervisors / electricians who are involved in inspection and testing of Electrical Installations

Course Duration

This course duration will be 4 days plus an evening examination.

Certification

Delegates will receive a City & Guilds certificate on successful completion of practical assessments and an external examination set by the awarding body.

Portable Appliance Testing
City & Guilds 2377- 200
Course Reference: CP - 8137

Aim
The aim of the course is to ensure that personnel are familiar with the use of a Portable Appliance Testing Machine and can apply it safely and correctly to test portable appliances.

Pre-Requisite
A knowledge of electricity/electrical safety would be an advantage.

Course Objectives
On completion of the course the delegate will be able to:

- Understand the need for portable appliance testing.
- Reasonably assess the required test period of a range of appliances.
- Distinguish between classes of appliances.
- Use a portable appliance tester, safely and accurately.
- Know when and how to perform a 'flash' test.
- Establish and maintain an accurate record of appliances, testing frequencies and a test history.
- Recognise and know how to overcome a range of problems peculiar to portable appliance testing.

After successful completion of this course further on-site training with an experienced PAT operative will be required to achieve competence

Course Duration
The course duration will be 1 days including City & Guilds assessments

Certification
Delegates will receive a City & Guilds certificate on successfully completion of the external examination.

Portable Appliance Testing (2377- 100)
Certificate of Competent Management of Electrical Maintenance

Course Reference: CP - 8203

Aim

The aim of the course is to ensure that personnel are familiar with the Codes of Practice and Legal aspects of the Management of an Electrical Equipment Maintenance Section.

Pre-Requisite

A knowledge of electricity/electrical safety would be an advantage.

Course Objectives

On completion of the course the delegate will be able to:-

- Understand the requirements of the Laws relating to electrical equipment.
- Classify types of electrical equipment identified in the Code of Practice.
- Identify the categories of inspection and test.
- Identify and record relevant information and interpret test results.
- State the training requirements for users of equipment and management of systems.
- List instrument types suitable for testing equipment and state testing procedures and parameters.

Course Duration

The course duration is one day.

Certification

Delegates will receive a City & Guilds certificate on successfully completion of the external examination.




ZONE 0



CompEx Hazardous Areas
Explosive Gas Atmospheres
Course Reference: CP - 8201

Aim
To provide Training and Assessment of Competence for electrical and instrument personnel for work on electrical apparatus for use in potentially explosive gas atmospheres

Pre-Requisite
Delegates must have a recognised Electrical or Instrument background with practical experience of flammable atmosphere installations.

Course Objectives
On successful completion of the course assessment the delegate will demonstrate competency in:-

- Ex01 - Preparation and Installation of Ex 'd' 'e' 'n' and 'p' systems in potentially explosive atmospheres.

- Ex02 - Inspection and Maintenance of Ex 'd' 'e' 'n' and 'p' systems in potentially explosive atmospheres

.

- Ex03 - Preparation and Installation of Intrinsically Safe Circuits in potentially explosive gas atmospheres.

- Ex04 - Inspection and Maintenance of Intrinsically Safe Circuits in potentially explosive gas atmospheres.

Course Duration
The course duration will be 5 days

Certification
On successful completion of the course assessments the delegate will receive a 'Certificate of Core Competency' from JT Ltd
The CompEx Certificate of Core Competence and renewal are valid for 5 years from the date of issue. These must be refreshed before expiry or the main CompEx assessments will have to be retaken.

CompEx Hazardous Areas Refresher and Update

Course Reference: CP - 8201R

Aim

This course is for personnel who wish to update and refresh their knowledge and skills in the use of electrical and instrument apparatus within potentially explosive gas atmospheres

Pre-Requisite

Candidates must hold a valid CompEx certificate of core competency.

Course Objectives

On successful completion of the course assessment the delegate will demonstrate competency in:-

- Ex01R - Installation Practices for Ex ‘d’ ‘e’ ‘n’ and ‘p’ apparatus
- Ex02R - Inspection and Maintenance for Ex ‘d’ ‘e’ ‘n’ and ‘p’ apparatus
- Ex03R - Installation practices for Intrinsically Safe Circuits
- Ex04R - Inspection and Maintenance for Intrinsically Safe Circuits

Course Duration

The course duration will be 2 days.

Certification

On successful completion of the course assessment the delegate will receive a renewal of the ‘Certificate of Core Competency’ from JT Ltd

The CompEx Certificate of Core Competence and renewal are valid for 5 years from the date of issue. These must be refreshed before expiry or the main CompEx assessments will have to be retaken.

CompEx Hazardous Areas
Combustible Dusts
Course Reference: CP - 8201D

Aim

To provide Training and Assessment of Competence for electrical and instrument personnel for work on electrical apparatus for use in potentially explosive atmospheres containing combustible dust.

Course Objectives

On successful completion of the course assessment the delegate will be awarded by JT Ltd a certificate of competency as follows:-

- EX05 - Preparation and Installation of Electrical Apparatus Protected by Enclosure for use in the Presence of Combustible Dusts.
- EX06 - Inspection and Maintenance of Electrical Apparatus Protected by Enclosure for use in the Presence of Combustible Dusts.

Course Duration

The course duration is 3 days. If the candidate has completed CompEx training units Ex01 - Ex04 within the previous 12 months then the course duration will be 2 days.

Certification

On successful completion of the course assessment delegates will be awarded a certificate of competency from JT Ltd

The CompEx Certificate of Core Competence is valid for 5 years from the date of issue and must be refreshed

CompEx Hazardous Areas Foundation

Course Reference: CP - 8201F

Aim

This course is intended primarily to give personnel an understanding of the requirements of safe working in a potentially explosive atmosphere. The course has been designed specifically for non-technical personnel, supervisors, contract managers and senior management. It is intended to provide a greater appreciation of CompEx core competency and the promotion of safe working or as an introduction to the main CompEx courses.

Course Objectives

On completion of the course delegates will have an: -

- Understanding of the current directives and regulations relating to hazardous areas
- Awareness of typical flammable atmospheres and forms of release
- Awareness of various ignition sources and potential consequences
- Understanding of the characteristics of gases, vapours and dusts
- Understanding of the basic principles of area classfication including zone, apparatus groups and temperature classification
- Awareness of the types of explosion protection for equipment for use in hazardous areas
- Understanding of the IP code for ingress protection
- Understanding of identification for hazardous area equipment

Course Duration

This course duration will be a minimum of 2 days.

Certification

On successful completion of the course assessments the delegate will be awarded a Certificate of Competence by JT Ltd

Achievement Measurement (AM2) Refresher

Course Reference CP - 8210

Aim

To prepare candidates with the necessary Refresher Training, Skills and Knowledge to sit the AM2 Skill Test

Pre- Requisite

Delegates must be in an electrical apprenticeship or undertaking an electrical qualification which requires the AM2 Skill Test

Course Objectives

During and on completion of this course the delegate will be able to:-

- Carry out suitable safe isolation techniques to the standard required by the EAW Regs 1989
- Follow suitable inspection and testing techniques for the requirements of the AM2 test
- Apply a Logical Approach to Fault Finding
- Have the theoretically understanding of various types of circuits
- Practise the practical aspects that will be assessed in the AM2 test

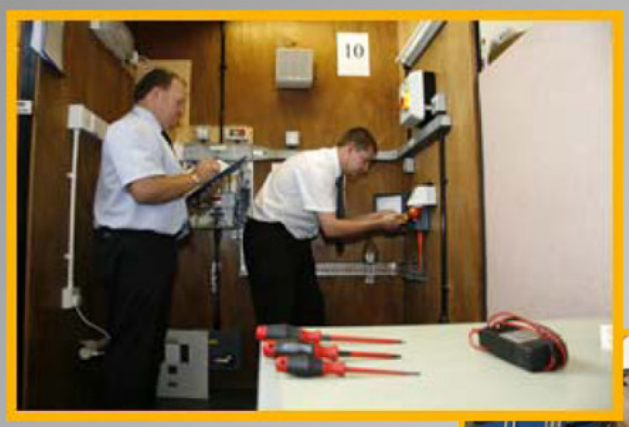
Course Duration

This course duration will be 2 days.

Certification

There is no certificate attached to this course

For details of the AM2 Skills Test
Please contact:-
Penny Dalby at NETA
on 01642 616936


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Instrument Skills Appreciation
Process Instrument Measurement & Control

Course Reference: CP - 9027

Aim

To enhance the skills and knowledge of personnel with regard to process instrument measurements and control

Pre- Requisite

Designed for personnel with a process or engineering background or instrument personnel requiring an update of skills

Course Objectives

On successful completion of the course delegate will be able to:-

- Understand the need to measure and control pressure, level, flow and temperature in the process industry to include all safety issues
- Understand the use of plant process & instrument diagrams to assist the fault location process
- Carry out all calibrations as applied to pressure, level, flow and temperature instrumentation using the correct test equipment 'on' and off' plant
- Understand the transmission of process measurement signals
- Understand control valve selection, operation and overhaul to meet plant operation and safety requirements
- Understand and demonstrate the use of automatic control in relation to 2 step through to 3 term control
- Understand the principles of DCS systems including configuration of a measurement & control loop.

Course Duration

The duration of the course is 10 days

Certification

Delegates will receive a NETA certificate on successful completion of the course

Instrument Skills Appreciation
Process Measurement
Course Reference: CP - 9027A

Aim

To enhance the skills and knowledge of personnel with regard to process instrument measurements

Pre- Requisite

This course is designed for personnel with a process or engineering background or instrument personnel requiring an update of skills.

Course Objectives

On completion of this course the delegate will be able to:

- Understand the need to measure pressure, level, flow and temperature in the process industry to include all safety issues
- Understand the use of plant process & instrument diagrams to assist the fault location process
- Carryout all calibration as applied to pressure, level, flow and temperature instrumentation using the correct test equipment 'on' and 'off' plant
- Understand the transmission of process measurement signals
- Appreciate process measurements with respect to process control
- Understand the priciples of Bus Systems measurements

Course Duration

The course is of five days duration.

Certification

Delegates will receive a NETA certificate on successful completion of the course

Instrument Skills Appreciation
Process Instrument Installation

Course Reference: CP - 9026

Aim

To provide a basic understanding and practical experience of Instrumentation installation techniques for the process industry.

Pre- Requisite

Candidates should have a process or engineering background

Objectives

On successful completion of the course the delegate will be able to: -

- Understand the need for instrumentation and what it achieves in relation to the process
- Understand and recognise British Standard instrument symbols as applied to Process & Instrument Diagrams (P&ID) and produce a P&ID for a simple process installation
- Appreciate how to measure and control process pressures, levels, flows and temperatures on 'live' plant
- Appreciate the 'control loop' and what it consists of in relation to the process and instrumentation
- Understand different types of fittings and pipe work used and their maximum working pressure
- Form and install stainless steel, copper and polythene tubing using correct fittings and support systems for a process instrument installation
- Correctly install a liquid pressure system correcting for any 'head' errors and commission the system
- Successfully complete an installation project from planning to commissioning on the training plant

Course Duration

The course is of ten days duration.

Certification

Delegates will receive a NETA certificate on successful completion of the course

Instrument Skills Appreciation in Process Control

Course Reference: CP - 9207B

Aim

To enhance the skills and knowledge of personnel with regard to process instrument control

Target Audience

This course has been designed for personnel with a process or engineering background or instrument personnel requiring an update of skills

Pre-Requisite

Delegates should have an understanding of Instrument process measurement

Course Objectives

On completion of the course delegate will be able to:-

- Confirm understanding of the need to measure and control pressure, level, flow and temperature in the process industry to include all safety issues
- Confirm understanding of the need to transmit representative signals of the process medium
- Understand control valve selection, operation and overhaul to meet plant operation and safety requirements
- Understand and demonstrate the use of automatic control in relation to 2 step through to 3 term control
- Understand the principles of DCS systems including configuration of a measurement & control loop

Course Duration

The duration of the course is five days

Certification

Delegates will receive a NETA certificate on successful completion of the course

Distributed Control System Course

Course Reference: CP - 9120

Aim

To give an appreciation of and a practical introduction to a Distributed Control System measuring and controlling Pressure, Level and Flow Loops on "live" plant.

To give an appreciation of how Analogue Measurement and Control Loops are configured and simple process graphics are created.

Pre-Requisite

The Course is aimed at "operator key" level for engineering and production personnel who require a basic understanding of the operation of a distributed control system. A knowledge of process operation would be required.

Course Objectives

On successful completion of this course delegates will be able to:-

- Understand DCS in relation to process control
- Demonstrate keyboard familiarity with the system and the different levels of use (operator, supervisor, configurer)
- Check analogue inputs and outputs both from the plant and at the field station
- Demonstrate the ability to control a simple process plant continuously.
- Correctly use a process graphics package to create and display process loops
- Apply the use of trend displays to evaluate control problems and adjust accordingly

Course Duration

The course is of two days duration.

Certification

Delegates will receive a NETA certificate on successful completion of the course

These Courses are one of many provided by NETA Training Group.

Other Courses include:

Scaffolding
Mechanical/Fabrication
Welding/Welder Testing
Lifting/Slinging
Pneumatics and Hydraulics
Management/Supervisory
Assessor Training A1/ A2/ V1
Health & Safety
Commercial & Domestic Gas Training
Transport
ADR Hazchem
DGSA
Lorry Loader
CPC (Certificate of Professional Competence)

For information on forthcoming
Courses Examination Dates / Times
or a full range of courses contact:

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Pennine Avenue
North Tees Industrial Estate
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TS18 2RJ
Tel: 01642 616936
Fax: 01642 612431
email enquiries@neta.co.uk website www.neta.co.uk

Meeting the needs and expectations of service users is of the highest priority to our staff. Therefore if you have any concerns or wish to make a comment about the service please contact: Barbara Seddon email ***b.seddon@neta.co.uk***

Equal Opportunities Statement

It is our aim that there shall be equal opportunities in this organisation. There will be no discrimination on grounds of sex, being married, colour, race or disability which is not permitted by law; or on any grounds, including age, except where this is necessary to ensure that the job is done effectively. This principle will apply to recruitment, training, promotion, dismissal, transfer and all other benefits, terms and conditions of employment.

16/10/2009 This information was correct at the date of publication